Solar ESS Co. Ltd.
53 Moo 2 Tambon Nanglae, Amphoe Muang, Chiang Rai 57100
Tel 063 151 6694 · 089 632 5829 · LINE solaress1
www.solaress.org · www.โซลาร์เอสส์.com

PLANT REVIEW · OFF-GRID FARM · NORTHERN CHIANG MAI

Healthy system, but one string of 6 panels is too small for cloudy, high-use days

Off-grid · 6 panels in one string (about 3.5 kWp) · Solis S6-EH1P6K-L-PLUS 6 kW hybrid · about 20 kWh battery · data 19 August–30 October 2026 · Rev 02 · 30 October 2026

6 panels
one string on MPPT1; MPPT2 is empty by design

3.4 kW
highest solar power recorded, normal for about 3.5 kWp

3 h 25 min
without power on 27 October (04:55–08:20), battery at its 20% cut-off

about 14 kWh
the most one string can make on a clear day; use reached 10.4 kWh on 26 October

Summary

Equipment is healthy

The string of 6 panels gives about 2.5 kW at noon on a clear day, normal for about 3.5 kWp. The inverter has no alarms and the battery cycles every day.

Too little solar for bad days

On cloudy days the string makes only 3–7 kWh. After a cloudy 26 October with 10.4 kWh of use, the battery reached its 20% cut-off at 04:55 and the farm had no power until 08:20.

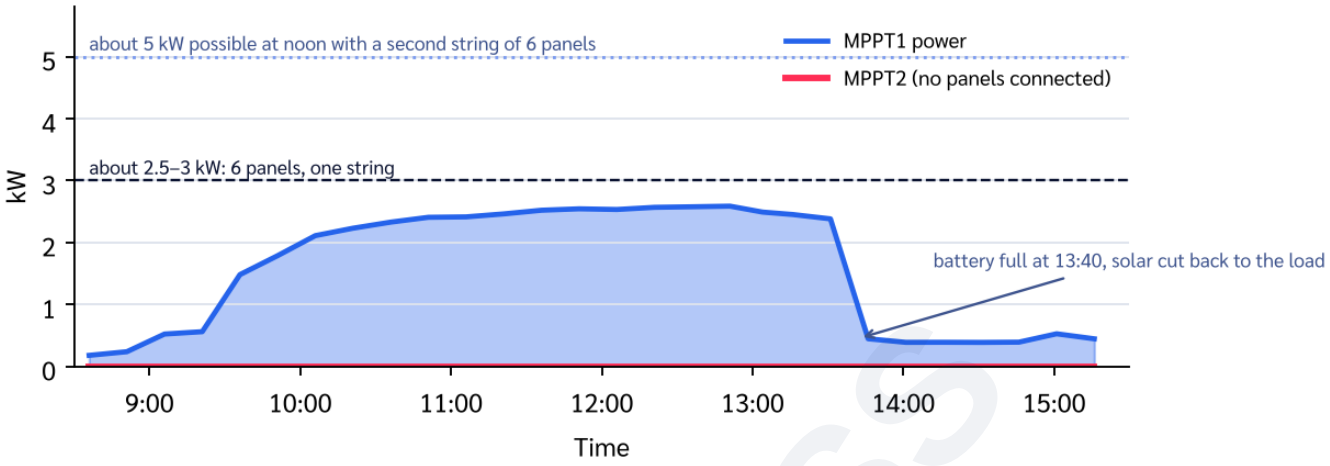
The fix: a second string

The inverter's second input (MPPT2) is free. Adding 6 more panels there would double solar and keep the battery charged through cloudy spells.

1 The system

Site	Off-grid farm, northern Chiang Mai
Type	Off-grid; all loads on the inverter backup port, no grid connection
Inverter	Solis S6-EH1P6K-L-PLUS 6 kW single-phase low-voltage hybrid, 2 MPPT inputs, commissioned 19 August 2026, no alarms
Solar array	6 panels in one string on MPPT1, about 3.5 kWp (estimated from voltage and power). MPPT2 is not used. SolisCloud lists 6 kWp and should be corrected.
Battery	Two low-voltage LFP batteries, about 20 kWh in total (8 kWh charged = 61% → 100%); maximum charge current allowed by the battery's management system 200 A; discharge cut-off 20%

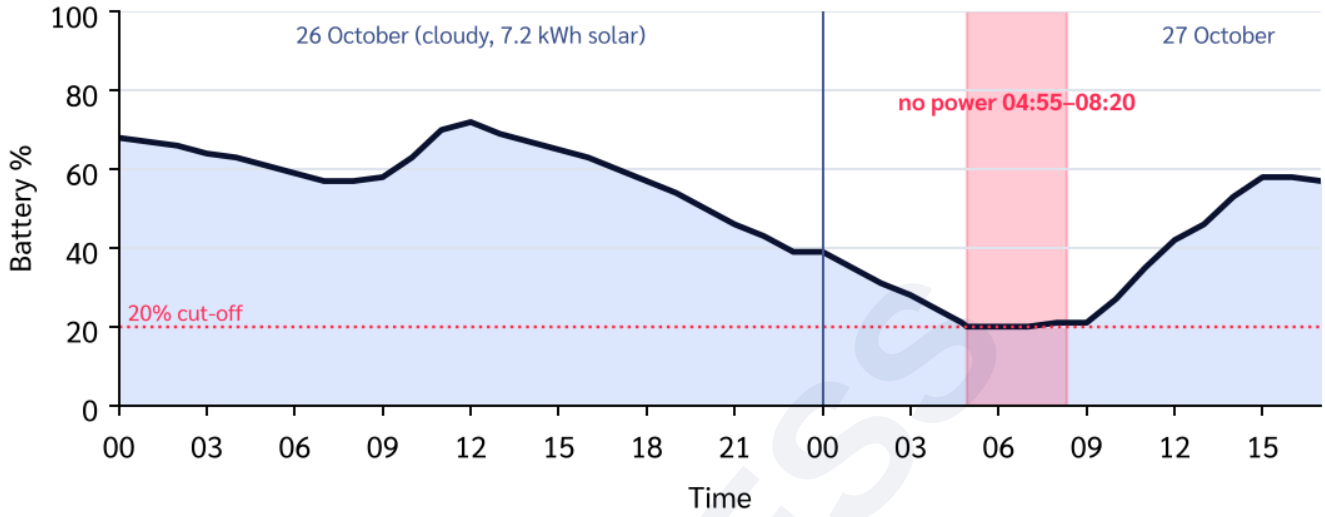
2 Solar input: one string of 6 panels, working normally



Date	MPPT1 max voltage	MPPT1 max power	MPPT2 voltage	MPPT2 current
25 August	260 V	1,734 W	30.7 V	**0**
15 September	281 V	2,935 W	30.7 V	**0**
10 October	293 V	2,791 W	30.2 V	**0**
27 October	281 V	3,377 W	29.9 V	**0**
30 October (clear)	281 V	2,620 W	30.5 V	**0**

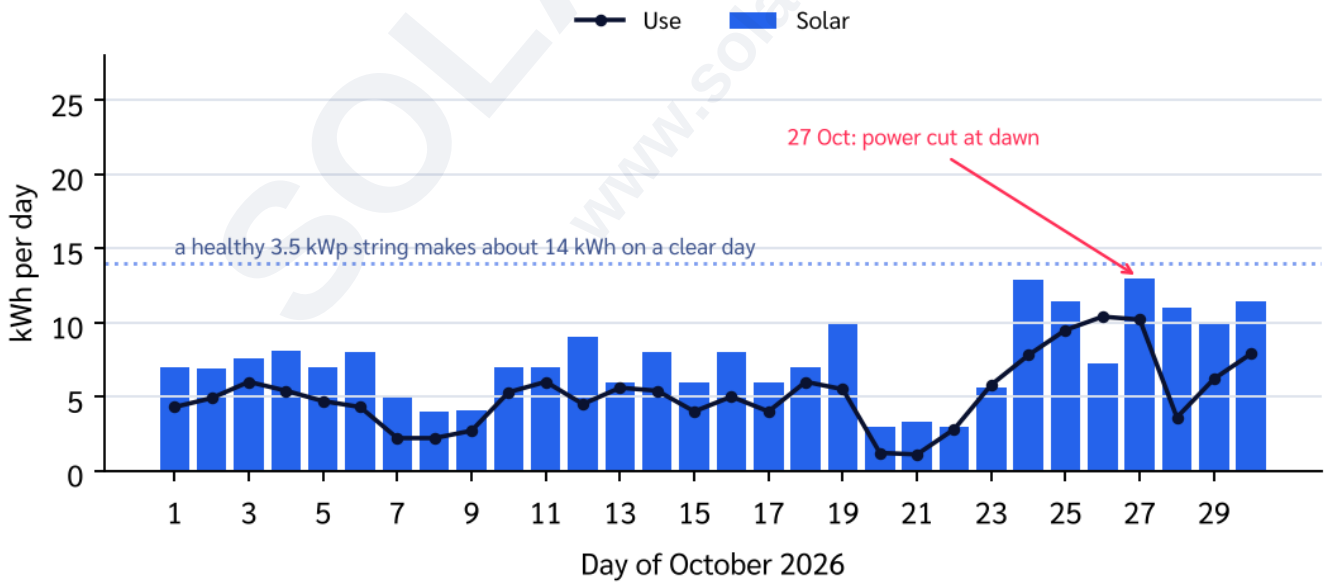
What the readings mean. MPPT1 shows 230–290 V and up to 14.5 A, which is normal for a string of 6 panels of about 3.5 kWp. On the clear morning of 30 October it gave a steady 2.4–2.6 kW, as expected at hot panel temperatures. MPPT2 reads 25–30 V and 0 A, the normal reading for an input with nothing connected. The battery is not a limit either: its management system allows 200 A of charging.

3 The power cut on 27 October



What happened. 26 October was cloudy: 7.2 kWh of solar against 10.4 kWh of use. The battery only reached 72% at midday, and the night load of about 0.7 kW drained it to the 20% cut-off at 04:55. The inverter switched the backup output off until the sun had recharged the battery enough, at 08:20. With a second string of 6 panels, 26 October would have produced about 14 kWh, enough to fill the battery and get through the night.

4 Production and use, August–October 2026



Month	Solar kWh	kWh/day	kWh/kWp/day	Use kWh
August*	83	6.4	1.83	54
September	188	6.3	1.79	126
October	224	7.5	2.13	158

* August from 19 August (commissioning). kWh/kWp is calculated on about 3.5 kWp (6 panels). Solar is also cut back on purpose once the battery is full, because there is no grid to export to.

5 Findings

1. **The string of 6 panels works normally:** about 2.5 kW at noon on a clear day and up to 13 kWh on a good day.
2. **MPPT2 is empty by design,** and its 0 A reading is normal. SolisCloud still lists the plant as 6 kWp.
3. **On sunny days the battery is full by about 13:30,** because use is low (about 5 kWh/day). After that, solar is cut back to the load.
4. **On cloudy days, or when use rises above about 8–10 kWh/day, one string cannot refill the battery.** On 27 October the farm was without power from 04:55 to 08:20.
5. **The inverter and battery are healthy.** There are no alarms, the battery charges and discharges every day, and its management system allows up to 200 A.

6 How to fix it

1. **Add a second string of 6 panels on MPPT2** Investment
   The inverter's second input is free. A matching string of 6 panels would roughly double solar to about 7 kWp and fill the battery even on cloudy days. Use the same panel model, or check that a different model fits MPPT2's voltage and current limits, and add its own DC isolator and fuse.
2. **Correct the capacity in SolisCloud** No cost
   Change the registered 6 kWp to the real array size (6 × panel wattage), so that yield figures and reports are correct.
3. **Watch the battery at dusk on cloudy days** No cost
   SolisCloud has no alert for battery level. Check the app in the evening after a cloudy day and reduce non-essential night loads if the battery is below about 40%.
4. **Run flexible loads between 11:00 and 15:00** No cost
   Pumps, washing and water heating use the surplus once the battery is full.
5. **Consider a generator for long rainy spells** Optional
   The inverter has a generator input. A small generator would cover several cloudy days in a row.
6. **Keep the 20% cut-off** No cost
   It protects the battery. The fix is more solar, not a deeper discharge.

7 Checklist before adding a second string

1. Panel model and wattage of the existing 6 panels; use the same model if possible.
2. Space for 6 more panels with the same tilt and direction, free of shade.
3. String voltage and current within MPPT2's limits on the coldest morning and hottest afternoon.
4. A separate DC isolator and string fuse for string 2, and correctly sized cable.
5. After connecting: MPPT2 current in SolisCloud above 0 A in daylight, and about 5 kW in total at noon on a clear day.

> **In one sentence:** the system is healthy, but one string of 6 panels cannot carry the farm through cloudy, high-use days. A second string on the free MPPT2 input is the most effective upgrade.

References

1. SolisCloud — the farm plant: daily, monthly and 5-minute inverter, MPPT and battery data, downloaded 30 October 2026
   Monitoring data
2. Solis S6-EH1P(3-6)K-L-PLUS single-phase low-voltage hybrid inverter — product page and datasheet
   https://www.solisinverters.com
3. SolisCloud push settings — available alerts: inverter offline, inefficient plant, string abnormal warning
   SolisCloud account settings

Figures come from SolisCloud monitoring data. Array size and battery capacity are estimated from the readings.